CHAPTER 13

OSCILLATIONS

CHAPTER ESSENCE AT A GLANCE:
Back and forth, back and forth! In Simple Harmonic Motion (SHM), the restoring force always pulls the particle back toward the center with force proportional to displacement ($a = -\omega^2 x$). Pendulums and springs tick with constant, dependable time periods!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Defining Condition of SHM:** Acceleration is directly proportional to displacement and opposite in direction: $a = -\omega^2 x$ (Restoring force $F = -k x$).

• **Displacement, Velocity & Acceleration:** Displacement $x(t) = A \cos(\omega t + \phi)$; Velocity $v = -\omega A \sin(\omega t + \phi) = \omega \sqrt{A^2 - x^2}$; Acceleration $a = -\omega^2 x$. Phase difference between x and a is π rad (180°)!

• **Energy in SHM:** Kinetic Energy $K = \frac{1}{2} k (A^2 - x^2)$; Potential Energy $U = \frac{1}{2} k x^2$; Total Energy $E = K + U = \frac{1}{2} k A^2$ = Constant. At mean position (x=0), all energy is KE; at extremes (x=±A), all energy is PE.

• **Simple Pendulum:** Time period $T = 2\pi \sqrt{L/g}$ (valid for small angles). T is INDEPENDENT of the mass or material of the bob! In a spaceship (g=0), $T \to \infty$ (pendulum clock stops!).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$x(t) = A\cos(\omega t + \phi), \quad v(t) = \pm\omega\sqrt{A^2 - x^2}$$

Displacement and Velocity in Simple Harmonic Motion

$$a(t) = -\omega^2 x(t), \quad F = -m\omega^2 x = -kx$$

Acceleration and Restoring Force in SHM

$$T = 2\pi\sqrt{\frac{m}{k}}, \quad T = 2\pi\sqrt{\frac{l}{g}}$$

Periods of Spring-Mass System and Simple Pendulum

$$E_{\text{total}} = K + U = \tfrac{1}{2} m\omega^2 A^2$$

Total Mechanical Energy Conservation in SHM

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The time period of a simple pendulum does NOT depend on the mass of the bob! A 1 kg bob and a 10 kg bob swing with the exact same time period.

• **Alert:** In SHM, Total Mechanical Energy $E = \frac{1}{2} m \omega^2 A^2$ is strictly constant! At the center, all energy is KE; at the extremes, all energy is PE.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Write the displacement equation of SHM x(t) = A cos(ω t + ϕ). State the expressions for maximum velocity and maximum acceleration. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Velocity and Acceleration]** Velocity v(t) = dx divided by dt = -ω A sin(ω t + ϕ). Max velocity v max = ω A (at mean position x = 0). [1 Mark]

• **[Step 2: Acceleration]** Acceleration a(t) = dv divided by dt = $-\omega^2$ A cos(ω t + ϕ) = $-\omega^2$ x. Max acceleration a max = ω^2 A (at extreme positions x = ± A). [1 Mark]

• **Key Exam Result:** v max = ω A, a max = ω^2 A

2. What happens to the time period of a simple pendulum if: (a) Its length is increased four times? (b) It is taken to the Moon (where g m = g divided by 6)? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Quadrupling Length]** Time period T = 2 π sqrt(L divided by g). (a) If L' = 4 L: T' = 2 π sqrt(4 L divided by g) = 2 x [2 π sqrt(L divided by g)] = 2 T (Time period doubles). [1 Mark]

• **[Step 2: Taken to Moon]** (b) On the Moon with g' = g divided by 6: T moon = 2 π sqrt(L divided by (g divided by 6)) = sqrt(6) T approx. 2.45 T (Time period increases by factor sqrt(6)). [1 Mark]

• **Key Exam Result:** (a) Time period doubles (2T); (b) Increases by factor sqrt(6)

3. Derive expressions for Kinetic Energy and Potential Energy of a particle executing SHM. Prove that total mechanical energy is conserved. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Potential Energy]** Restoring force F = -k x. Work done dW = k x dx. U = $\int_0^x$ k x dx = ½ k x^2 = ½ m ω^2 x^2. [1 Mark]

• **[Step 2: Kinetic Energy]** Speed v = ω sqrt(A^2 - x^2). Kinetic energy K = ½ m v^2 = ½ m ω^2 (A^2 - x^2). [1 Mark]

• **[Step 3: Total Energy Conservation]** Total energy E = K + U = ½ m ω^2 (A^2 - x^2) + ½ m ω^2 x^2 = ½ m ω^2 A^2 = ½ k A^2. Total mechanical energy is constant, independent of position x and time t. [1 Mark]

• **Key Exam Result:** E total = ½ m ω^2 A^2 = Constant at all points

4. Derive an expression for the time period of small oscillations of a Simple Pendulum: T = 2 π sqrt(L divided by g). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Restoring Force Resolution]** For a simple pendulum of length L and bob mass m displaced by small angle θ: Forces acting are tension T str along string and gravity mg downward. Tangential restoring force F = -mg sin θ. [1 Mark]

• **[Step 2: Small Angle Approximation]** For small oscillations (θ << 1 rad), sin θ ≈ θ = x divided by L (where x is arc displacement): F = -mg (x divided by L) = -(mg / L) x. Restoring acceleration a = $\frac{F}{m}$ = - $\frac{g}{L}$ x. [1 Mark]

• **[Step 3: Time Period Formulation]** Comparing with standard SHM equation a = $-\omega^2$ x: ω^2 = $\frac{g}{L}$ ⇒ ω = $\sqrt{\frac{g}{L}}$. Time period T = 2 π divided by ω = 2 π √(L divided by g). [1 Mark]

• **Key Exam Result:** Time Period T = 2 π sqrt(L divided by g)

5. Derive an expression for the time period of vertical oscillations of a mass m suspended from a light helical spring of force constant k: T = 2 π sqrt(m divided by k). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Equilibrium Position]** In equilibrium with extension l: k l = m g. [1 Mark]

• **[Step 2: Additional Displacement]** When pulled down further by distance y: Net restoring force F = mg - k (l + y) = mg - kl - ky = -ky. Restoring acceleration a = $\frac{F}{m}$ = -(k divided by m) y. [1 Mark]

• **[Step 3: Time Period]** Comparing with a = $-\omega^2$ y: ω^2 = k divided by m => ω = sqrt(k divided by m). Time period T = 2 π divided by ω = 2 π sqrt(m divided by k). [1 Mark]

• **Key Exam Result:** T = 2 π sqrt(m divided by k)

6. A particle executes SHM of amplitude 5 cm and period 0.2 s. Find its: (a) Maximum velocity (b) Maximum acceleration (c) Velocity when displacement is 3 cm. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Angular Frequency]** Amplitude A = 5 cm = 0.05 m, period T = 0.2 s $\Rightarrow \omega = 2\pi$ divided by T = 2π divided by 0.2 = 10π rad/s $\approx$ 31.42 rad/s. [0.5 Mark]

• **[Step 2: Maximum Velocity and Acceleration]** v max = ω A = (10π) x 0.05 = 0.5π m/s $\approx$ 1.57 m/s. a max = ω^2 A = $(10\pi)^2 \times 0.05 = 100\pi^2 \times 0.05 = 5\pi^2$ m/s^2 $\approx$ 49.35 m/s^2. [1.5 Marks]

• **[Step 3: Velocity at x = 3 cm]** At x = 3 cm: $v = \omega\sqrt{(A^2 - x^2)} = (10\pi)\sqrt{(5^2 - 3^2)} = (10\pi)\sqrt{(16)} = 40\pi$ cm/s = 0.4π m/s $\approx$ 1.26 m/s. [1 Mark]

• **Key Exam Result:** v max = 1.57 m/s, a max = 49.35 m/s^2, v(3cm) = 1.26 m/s

7. At what displacement from the mean position is the kinetic energy of a particle in SHM equal to its potential energy? **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Energy Formulations]** Kinetic energy K = ½ k ($A^2 - x^2$); Potential energy U = ½ k x^2. [1 Mark]

• **[Step 2: Equating Energies]** Given K = U: ½ k ($A^2 - x^2$) = ½ k x^2 => $A^2 - x^2 = x^2$ => 2 $x^2 = A^2$. [1 Mark]

• **[Step 3: Solving for x]** $x^2 = A^2$ divided by 2 => x = ± A divided by sqrt(2) approx. ± 0.707 A. [1 Mark]

• **Key Exam Result:** Displacement x = ± A divided by sqrt(2) approx. ± 0.707 A